ModelIt!

Student Activity Pack

Why Are There So Many Kinds of Bugs?

Exploring the Diversity of Life on Earth

Grade 2 | NGSS 2-LS4-1 | Structure and Function

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 How many different kinds of bugs can you name? Why do you think there are so many?

2 Do you think a butterfly and a beetle need the same body features? Why or why not?

3 Draw your favorite insect and label the body parts that you think help it survive.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Habitat Type | Body Features | Survival Success

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Garden Habitat

Habitat Type set to garden

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Pond Habitat

Habitat Type set to pond

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Forest Floor

Habitat Type set to dark forest floor

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Predator Type — The kind of animals that eat the insect — birds, frogs, or bigger insects. Different predators require different defenses.

Food Source — What the insect eats in its habitat — nectar, leaves, other insects, or decaying matter. Different foods need different mouthparts.

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why are there so many different kinds of bugs — and why do they all look so different from each other? Use evidence from your simulation to support your answers.

1 Why do insects in a pond look so different from insects in a garden?

2 If all insects had the exact same body, could they all survive in every habitat? Why or why not?

8 STEM Challenge: Design a Bug for a New Habitat

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YOUR ENGINEERING MISSION

Design an imaginary insect with body features perfectly suited to survive in a habitat you choose.

THE SCENARIO

Scientists have discovered a brand-new habitat — and they need your help to predict what kind of insect could survive there! Your team will choose a habitat (real or imaginary) and design an insect with body features that help it find food, escape danger, and thrive.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE TO BUILD YOUR INSECT MODEL — CLAY, PIPE CLEANERS, PAPER, OR RECYCLED MATERIALS?

2. HOW MANY LEGS, WINGS, AND SPECIAL FEATURES DOES YOUR INSECT HAVE?

3. WHAT COLORS AND PATTERNS WILL YOU GIVE IT AND WHY?

4. HOW WILL YOU PRESENT YOUR INSECT TO THE CLASS AND EXPLAIN WHY IT IS PERFECT FOR ITS HABITAT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Entomologists are scientists who study insects — the most diverse group of animals on Earth! They discover new species, protect crops from pests, and study how insects help ecosystems. They earn $45,000–$85,000/year.